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A Multi-Stage Prompting and RAG Approach to Generating Legal Analysis in Common Law Systems

Titus T. H. NG^a, Tien-Hsuan WU^a, Benjamin Minhao CHEN^a, Yongxi CHEN^b, and Ben KAO^a

^a*The University of Hong Kong*

^b*Australian National University*

Abstract. Retrieval-augmented generation (RAG) has shown success in many knowledge-intensive natural language understanding tasks through provision of external knowledge to large language models (LLM). We explore the use of multi-stage prompting in generating legal analysis. We break the complex legal problems into small steps, including initial analysis, legal query formulation, legal sources retrieval, legal reasoning, result consolidation, and output generation. The design of this framework emulates the process a legal professional in a common law system approaches the problem.

Keywords. legal analysis generation, multi-stage prompting, retrieval-augmented generation

1. Introduction

Rising legal costs present a significant barrier to justice, prompting exploration of artificial intelligence as a potential solution. However, large language models (LLMs) often struggle with complex legal reasoning in zero-shot scenarios. This research investigates how a multi-stage prompting and retrieval-augmented generation (RAG) approach, designed to imitate important features of the cognitive process of common law lawyers, can enhance AI-generated legal analysis in common law systems, specifically focusing on the legal framework of the Hong Kong Special Administrative Region.

Legal analysis demands a structured approach, typically involving the identification of legal issues, searching for applicable rules from the sources of law (including legislation and precedents), and careful application of the rules. Our study proposes a novel five-stage pipeline meticulously crafted to emulate the thought process and methodology of experienced common law practitioners: legal issue identification, legal research question formulation, legal source retrieval, deductive legal reasoning, and legal reasoning by analogy. This methodology aims to break down the complex task of legal analysis into manageable steps for the machine, imitating the systematic approach employed by human lawyers and potentially improving the relevance, accuracy and reliability of AI-generated results.

The research question at the core of this study is: “How does a multi-stage prompting and retrieval-augmented generation (RAG) pipeline, designed to emulate a common law lawyer’s approach, enhance the relevance and accuracy of LLM-generated reasoning results concerning the legal issues in the common law legal system?” By addressing this question, we seek to quantify the improvements offered by our proposed method and assess its potential as a tool for making AI-generated advice about legal issues more accessible and affordable.

In summary, this research aims to bridge the gap between expensive human legal expertise and the limitations of current AI systems in emulating human legal reasoning by closely modeling important features of the reasoning process of common law lawyers. By doing so, we potentially pave the way for more accessible and reliable AI-assisted legal advice in common law jurisdictions, while enabling the answer generation processes of LLMs to imitate significant aspects of the legal analytical approach.

2. Related Work

In common law systems, legal reasoning involves identifying legal norms from sources of law, synthesizing legal rules into a coherent framework, and applying the rule to the facts [1]. After identifying a rule of law, the rule can supply the major premise, while the facts form the minor premise, then a conclusion can be reached by applying the rule to the facts [2]. Argument schemes for reasoning by analogy includes arguments, rebuttal arguments, and surrebuttal arguments [3]. It has been argued that the judiciary should first use formal reasoning such as deductive reasoning, then proceed to reasoning by analogy if it is inadequate, and finally only proceed to policy arguments if the former two types of reasoning are still not adequate [4].

In the past decade, a shift has occurred in the field of AI and law from a “knowledge-based approach”, which is manually representing legal rules to enable computational models to mirror human reasoning and resolve legal problems, to a machine-centric approach, which is using machine learning to automatically identify patterns from legal texts and predict outcomes of legal problems [5]. The patterns generated by machines were often distinct from the elements of human reasoning process [6]. However, there have been increasing attempts to enable machine learning, especially large language models, to take into account the features of human reasoning and achieve superior results in reasoning tasks. Most notably, Wei, et al. [7] propose the concept of *chain-of-thought* prompting. Through providing some simple chain-of-thought demonstrations, the model could also describe the steps it takes to arrive at the answers. Gao, et al. [8] propose modular RAG to decompose complex queries and perform different RAG patterns such as conditional, branching, and looping. In our work, we investigate multi-stage prompting in the legal domain by mimicking the major steps that a legal professional would often go through when analyzing legal issues. The process is divided into multiple steps, each handled by the LLM with a different prompt.

3. Method

In this section, we describe the dataset and present our multi-stage prompting and retrieval-augmented generation (RAG) approach for generating legal reasoning.

We collected the ordinances and subsidiary legislation in Hong Kong and the judgments from the Hong Kong Court of Final Appeal (CFA), Court of Appeal (CA), and Court of First Instance (CFI) as the data to support our system. The data are kept in four databases: the legislation (LEGIS) database for the ordinances and subsidiary legislation, and three precedent databases (CFA, CA, CFI) for judgments. We processed the data and divided the legislation and the judgments into chunks. For legislation, it is typically structured according to the hierarchy: part $\rightarrow$ division $\rightarrow$ section $\rightarrow$ subsection. We partition the legislation at the section level. However, if a section is too long and cannot fit into one chunk (exceeding 512 tokens), we break the section in the subsection level, and use additional chunks to include the remaining subsections while keeping the structure of the section. For judgments, we divide each case into 1000-token chunks. We also empirically determine the overlap of each chunk to be 40% to better capture the context. For each chunk, we also store its BGE-M3 and BGE-Reranker-Gemma embeddings to facilitate RAG.

3.1. Multi-Stage Prompting and RAG

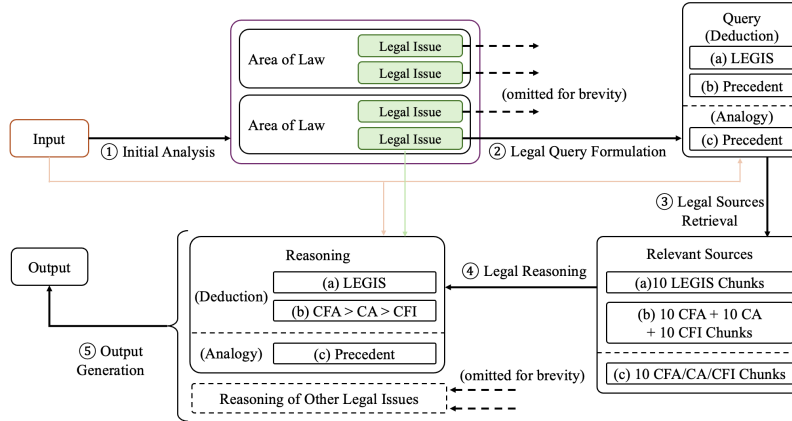


Figure 1. Framework of multi-stage prompting and RAG

Our multi-stage prompting system aims to emulate the reasoning process of a legal professional when answering a legal question. To achieve this goal, we break down the reasoning process into multiple steps, and design a module to handle each reasoning step. Figure 1 shows the diagram of our multi-stage prompting system. We will discuss each module in this section. In this work, we adopt GPT-4o as the LLM for demonstration purpose. We remark that it is possible to adopt other generative LLMs.

① *Initial Analysis* Given an input question or scenario, a legal professional would first analyze the question to identify the areas of law and the issues pertinent to the question before conducting a deeper study. The initial analysis is mostly based on the knowledge possessed by the legal professional without collecting other legal materials. In our approach, we provide a list of 8 civil law areas and prompt the LLM to identify the relevant ones. We also apply prompt learning by providing some examples in the prompts as it significantly improves the performance of LLM. The prompt for identifying legal areas

is omitted from the paper for brevity; the complete prompts, examples and parameters can be found in a separate document¹.

The identified legal areas guide the LLM in constructing legal issues. For each area, we further provide the table of contents of the legal practitioner's textbook in that area of law, and ask it to identify some legal issues. After identifying the areas of law, we further prompt the LLM to identify some potential legal issues for each area. For example, an issue could be "whether a company has a duty of care to ensure the elevator operates safely" or "whether the contract between two parties constituted a sale of specific goods or unascertained goods."

② *Legal Query Formulation* After identifying relevant areas of law and potential legal issues, the next step is to formulate the legal queries for conducting searches in legal information systems. We emulate the process by prompting the LLM to generate queries. For each legal issue, we generate several query keyphrases: (a) to query the legislation database, (b) for deductive reasoning with precedents, and (c) for reasoning by analogy with precedents. For (a) and (b), we prompt the LLM to generate a query for each legal issue and instruct the LLM that the query string shall encapsulate the semantic meaning of the question to be searched. For (c), we aim to retrieve cases containing facts similar to the user input. We prompt the LLM to generate key phrases to retrieve cases having different facts but sharing similar fact patterns. See footnote 1 for the complete prompt.

③ *Legal Sources Retrieval* This step emulates the process of conducting searches in legal information systems. Based on the query formulated in the previous step, we retrieve (a) 10 chunks of legislation from the legislation database (LEGIS), (b) 10 chunks of judgments most related to the legal issue from each of the precedent databases (CFA, CA and CFI), and (c) 10 chunks of judgments that contain similar fact patterns from the precedent databases. The retrieval is done in two steps. First, we use BGE-M3 embeddings to filter top 100 chunks similar to the query (measured by the cosine similarity) as candidates. Then we rerank the candidates using a larger embedding model, BGE-Reranker-Gemma, and finalize the result by selecting the top 10 related chunks.

④(a) *Legal Reasoning (Legislation; by Deduction)* After retrieving the related legislation and judgments, we perform a reasoning process for each retrieval result. For legislation, deductive reasoning is applied. We prompt the LLM to identify *major premises* (i.e., legal rule) mentioned in the retrieved chunks. Each major premise consists of four elements: the *conditions* of the rule, the *consequence* of the rule, the *exceptions* to the rule, and the *citation* from which the rule is extracted. If no major premise can be identified (i.e., *irrelevant*), we instruct the LLM to output an empty list. See footnote 1 for the complete prompts and the examples.

For each major premise, we then prompt the LLM to check (i) if the conditions of applying the premise is met in the current case, and (ii) if any condition makes the premise inapplicable in the current case. We ask the LLM to output the reasoning process (i.e., *minor premise*) and conclude which major premises apply to the current case.

④(b) *Legal Reasoning (Precedent; by Deduction)* Besides the LEGIS database, we also retrieve 10 relevant chunks from each of the three precedent databases. As judgments are long and complex, the retrieved judgments may be relevant to the legal issue but not to the

¹ <https://anonymous.4open.science/r/jurix2024-9/>

scenario input by the user. To address this issue, we instruct the LLM to select 2 chunks (among the 10) from each database that are related to the input and the legal scenario. For each selected chunk, we enrich its context to form the *excerpt* by concatenating it with 4,000 preceding and succeeding tokens in the same judgment. Similar to the deductive reasoning for LEGIS results, we prompt the model to identify major and minor premises given each excerpt (see footnote 1). As there is a hierarchy among the courts (CFA>CA>CFI), we start generating reasoning results using the cases from the highest court (CFA), and move down to lower courts only when there is no relevant legal rule.

④(c) *Legal Reasoning (Precedent; by Analogy)* Besides deductive reasoning, we also include reasoning by analogy to consider cases that have different facts but share similar fact patterns. From the 10 retrieved judgments, we further ask the LLM to pick one that is most relevant to the input scenario. After the judgment is picked, we follow the argument schemes for case-based analogies in [3] and instruct the LLM to perform the 11-step process listed below. See footnote 1 for the complete prompt.

- **Steps 1–4 Generating the Argument by Analogy:** Citing Precedent; Identifying Shared Legal Concepts; Proposing a Legal Rule; Applying the Rule to the Current Case.
- **Steps 5–8 Generating the Rebuttal Arguments:** Distinction Due to Missing Feature in Current Case; Argument from Undesirable Consequence; Distinction from Non-occurring Undesirable Consequence; Distinction from Missing Feature in Precedent Case.
- **Step 9&10 Generate the Surrebuttal Arguments:** Downplaying Significance of Distinction; Feature Substitution.
- **Step 11 Concluding the Argument:** Making Final Decision.

⑤ *Output Generation* The three reasoning results of each issue are concatenated as the output to the issue. To obtain the output w.r.t. the user input, we further concatenate the reasoning results of all legal issues.

4. Discussion and Conclusion

To evaluate the effectiveness of our multi-stage prompting legal reasoning approach, we collected 5 questions from a law exam preparation book [9]. We invited two legal experts to evaluate and comment on the responses.

For the responses generated by deductive reasoning (i.e., Step 4 (a)&(b)), the results are satisfactory. The reasoning can be generated with the relevant legislation and precedents retrieved. This could potentially help a lawyer prepare a case and retrieve legislation and precedents from the legal database. We also observe that our method generates fine-grained issues, some of which could possibly be worth exploring in a non-examination real-world setting where the universe of facts is not closed. For reasoning by analogy (i.e., Step 4 (c)), one limitation is that the exam prep questions tend to cover legal areas that are well-defined and the legal rules are relatively easy to identify. As such, the effectiveness of reasoning by analogy could not be properly reflected in the evaluation. In our future work, we will test the analogical reasoning with hypothetical cases on issues where legal rules are not well established. We will also design a mechanism to determine the optimal reasoning approach (deductive or analogical) given the user input.

In this paper, we proposed an innovative idea of multi-stage prompting and RAG approach to generate legal reasoning. The prompt in each stage is meticulously crafted

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to instruct the LLM to focus on a small and specific task. Through this approach, we could inject argumentative knowledge such that LLM knows the procedure to follow. By capturing important features of human legal reasoning, this approach also potentially increase the explainability of LLM-generated results for legal problems. The deductive reasoning has shown satisfactory results, and future work will focus on evaluating the reasoning by analogy and improving the overall reasoning responses.

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